

LLM 인용 방식의 재고찰: 단순 키워드 매칭의 가능성 탐색 (Rethinking LLM Citations: Exploring the Potential of Simple Keyword Matching)

Abstract

대규모 언어모델(LLM)의 활용이 증가함에 따라 환각(hallucination) 문제를 완화하고 신뢰성을 확보하기 위한 인용 기법의 중요성이 커지고 있다. 기존의 사후 인용 방법은 주로 임베딩 기반 유사도 계산에 의존하지만, 이는 핵심 키워드의 의미를 충분히 반영하지 못하고 계산 비용 또한 높은 한계를 가진다. 본 연구는 키워드 추출을 기반으로 문장 단위에서 인용을 수행하는 새로운 사후 인용 방법을 제안한다. 제안 방법은 생성 문장과 문서 간의 키워드 일치도를 활용하여 보다 정확하고 해석 가능한 인용을 제공하며, 추가적인 학습 없이도 빠르고 효율적으로 동작한다. 다양한 데이터셋과 평가 방법을 통해 기존 방법 대비 인용 정확도와 계산 효율성이 향상됨을 확인했다.

키워드: 키워드 매칭, 사후 인용, 검색증강생성, 대규모 언어 모델

Abstract

With the rapid advancement of Large Language Models (LLMs), improving their reliability and accountability has become increasingly important, particularly due to issues such as hallucinations. Retrieval-Augmented Generation (RAG) addresses this by grounding outputs in external knowledge, but accurately citing sources remains a challenge. Existing post-hoc citation methods typically rely on embedding-based similarity, which may fail to capture critical keyword-level nuances and often incur high computational costs. In this paper, we propose a novel keyword-centric post-hoc citation method that operates at the sentence level. By explicitly matching keywords between generated text and retrieved documents, the proposed approach improves citation accuracy while remaining interpretable and computationally efficient. Notably, it eliminates the need for training neural networks and significantly reduces resource requirements. Experimental results on multiple datasets, including the Automatic LLMs' Citation Evaluation-based evaluations and GPT-4 assessments, demonstrate that our method consistently outperforms conventional approaches in both accuracy and efficiency. These findings highlight the potential of simple keyword matching as a practical alternative for enhancing the reliability of LLM-generated content.

Keywords: Keyword matching, Post-hoc citation, RAG, LLM

1 Introduction

The rapid advancement of Large Language Models (LLMs) such as Llama [1], ChatGPT [2], and others, has marked a significant milestone in the field of Natural Language Processing (NLP). With the increasing use of LLMs in tasks such as Question Answering, which depends heavily on factual information, the need to enhance their reliability grows correspondingly. Precise and appropriate citations can boost credibility [3]. In many real-world domains—particularly in medical or scientific settings—it is crucial to link each generated statement to trustworthy external sources. However, existing post-hoc citation methods often rely on matching entire sentences to documents via embedding-based similarity, which can overlook domain-specific terms vital for accurate referencing. Moreover, the high computational cost and memory footprint of these methods pose practical limitations, especially when dealing with large-scale or resource-constrained environments.

To address these shortcomings, we propose a post-hoc citation methodology that not only improves citation accuracy but also operates over ten times faster than conventional methods while using fewer resources. This performance enhancement is attributed to our keyword-centric approach when utilizing Retrieval-Augmented Generation (RAG) [7]. By concentrating on specific keywords, this method streamlines the information, highlighting the essentials and eliminating unnecessary details. Additionally, our model utilizes less memory without the need for extensive training.

Fig. 1 presents an example question and answer from PubMedQA [8], to the best of our knowledge, highlighting the difference between our model and the conventional yet state-of-the-art method of similarity search [9] within the RAG framework. We selected PubMedQA because it consists of biomedical research questions with verifiable answers, making it well-suited for evaluating citation accuracy in a fact-heavy domain. We also used Medline, a curated medical literature database, to assess the method's effectiveness in retrieving accurate sources for technical queries. Both datasets contain domain-specific text requiring high-precision citations, aligning with the objective of our keyword-centric citation framework. Both methods are post-hoc citation methods. We used the top 15 documents related to the query, among which the four shown below were cited. Therefore, the sentences generated by the baseline method and our model are identical;

Question:

Can the growth rate of a gallbladder polyp predict neoplastic polyp?

Answer

Yes, the growth rate of a gallbladder polyp can sometimes predict its malignant potential.

Studies have shown that polyp growth rates can be used as a predictive factor for malignant transformation.

Specifically, a growth rate of 3 mm per year in polyp is a significant predictor of an increased risk of malignant transformation.

	Base	Ours
[1]	[1]	[2]
[1]	[1]	[3]
[1]	[1]	[3]

Documents

[1] Moerman and Bueno-de Mesquita [...] Probably at a higher risk are those who are exposed to stones for longer periods. An indicator of duration of exposure is not presently available; whether stone size can be such an indicator in specific conditions and populations needs [...]

[2] Diep et al. The growth rates of solitary gallbladder polyp compared to multi-polyps: A quantitative analysis. Several studies suggest solitary gallbladder polyp (GBPs) have an increased malignancy risk compared to multi-polyps. Furthermore, some malignant GBPs possessed faster growth rates compared to benign GBPs [...]

[3] Han et al. [...] In conclusion, our study showed that a GB polyp growth rate of 3 mm or more per year is a significant predictor of neoplastic polyps. In addition to the previously established risk factors for neoplastic GB polyps, cholecystectomy should be considered if the GB polyp growth rates is 3 mm/year or more.

Figure 1: Example question and answer with citations in medical domain compared with the baseline method (marked as ‘Base’; Baseline model as Maximum Inner Product Search) and our Keyword Matching method (marked as ‘Ours’)

the only difference lies in the citations. The documents at the bottom are articles from PubMed¹ and Medline² stored in our vector database for the experiment.

The baseline method citations were marked as [1], [1], [1] for the answer, while our method cited [2] for the first and [3] for the second sentence and the third sentence. The words highlighted in the same color are the overlapping keywords between the document and the sentence. The document [1] was not directly related to the answer, whereas [2] and [3] were relevant to it, indicating that our method achieved more accurate citations. Dense embeddings capture the overall meaning of a sentence, but they may overlook important keywords because they compress the sentence into a fixed-length representation. This can result in inaccurate citations as shown in Fig. 4. Conversely, our model identified keywords that commonly appear both in the document and the generated sentence, enabling

¹<https://pubmed.ncbi.nlm.nih.gov/>

²<https://medlineplus.gov/>

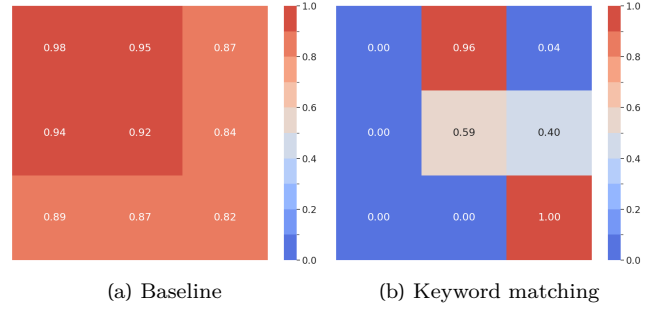


Figure 2: Heatmaps comparison: The x-axis is labeled from left to right as document 1, 2, 3, and the y-axis is labeled from top to bottom as sentence 1, 2, 3. The redder the color, the higher the similarity, and the bluer the color, the lower the similarity.

direct and relevant citations to the answer. The higher ratio of overlapping keywords in these two documents compared to others proposed the reason for a clearer interpretation of why they were cited.

As shown in Fig. 2, the baseline method generated a score matrix that is difficult to interpret when calculating the similarity between the sentences and the documents. This minimal variability highlights the method’s difficulty in effectively distinguishing between different types of content. Conversely, our framework showed a more distinct segregation of score distributions, thereby contributing to the improved performance of our method.

Our analysis of the baseline method’s inherent limitations, particularly evident from the self-attention analysis in the first sentence of Fig. 1, revealed a tendency to highlight less informative words such as “yes” and “predict”, while often overlooking more significant words. This pattern persisted across various documents, with a preference for words less pivotal for citation like “diagnosed” and “possessed” rather than focusing on key terms crucial for citations, such as “gallbladder” and “polyp”. This pattern suggests that the model’s embeddings are influenced by words that are not as relevant to the core information the citation aims to convey.

We hypothesize that this discrepancy is due to a kind of noise that distracts the model from focusing on the essential terms necessary for precise citations. For example, we infer that the encoder was a lack of understanding of medical domain terms when transforming the answers into dense embeddings for inner product calculation. This limitation could stem from biases inherent in the pre-trained knowledge of the LLM, potentially leading to improper embeddings. This hypothesis is supported by our observation that even when we alter these specific words to others, the model’s attention remains improperly fixated on these new words, despite their lack of importance for the citation process. By removing this noise, our approach was able to more accurately map the connections between documents and sentences, leading to an improvement in performance.

Fig. 2 visualizes the similarity matrix used in citation selection. The x-axis represents retrieved documents, while the y-axis represents sentences in the gen-

erated response. The color intensity indicates similarity, with redder regions representing higher similarity scores. Unlike the baseline method, which produces a uniformly distributed similarity matrix, our keyword matching approach results in clearer score differentiation, improving citation accuracy.

Moreover, as shown in Fig. 3, our model operates faster while requiring minimal resources. We checked the processing times of two essential steps for citation: loading the model and performing inference, excluding common processes such as sentence generation. In both steps, comparing our model to the conventional model [9], we observed $20.6\times$ decrease in processing time during inference and $17.9\times$ less memory usage comparing the ALCE framework which uses sentence transformer ‘gtr-t5-xxl’.³ Even among the same models, it demonstrated a much faster speed compared to the smaller-sized models.

In summary, our main research objective is to develop a resource-efficient post-hoc citation method that overcomes the drawbacks of existing dense embedding-based approaches. This design is motivated by an empirical observation reported in Sec. 3: across four LLMs and three datasets, keywords appearing in the retrieved documents are reproduced in the generated text at a consistently higher rate than non-keyword general nouns (e.g., 0.21 vs. 0.08 on ASQA), indicating that keyword retention is a usable signal for source attribution. By concentrating on key domain-relevant terms, our approach delivers more accurate, interpretable, and faster citations.

Key Contributions

- **Clarification of Limitations in Existing Methods:** We highlight how conventional post-hoc citation approaches can overlook critical keywords due to dense embedding compression.
- **Keyword-Centric Citation Framework:** We introduce a straightforward, keyword-focused method for sentence-level citation, circumventing the pitfalls of maximum inner product-based similarity.
- **Efficiency and Resource Reduction:** We empirically show that our method achieves high performance while operating ten times faster and using fewer computational resources, thereby broadening its applicability.

2 Related works

Our method is a keyword-centric post-hoc citation method with the RAG structure. Therefore, we will present an overview of related work on retrieval, citation, and keyword extraction.

³<https://huggingface.co/sentence-transformers/gtr-t5-xxl>

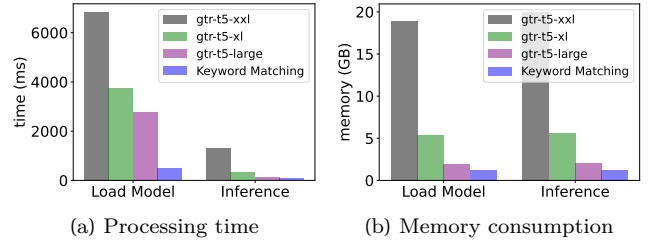


Figure 3: Comparison of processing time (left) and GPU memory consumption (right). All of the models were tested in the same environment (Intel Xeon Gold 6342 and Nvidia A6000). Models were loaded with full precision (FP32) and used batch size 32 for inference.

2.1 Knowledge Grounding through Retrieval in LLMs

Even widely used LLMs in various fields still face the issue of hallucination [10], so there has been research regarding the grounding with retrieval [11]. Lewis et al. [7] proposed utilizing a retriever for grounding external knowledge in its generation process. Karpukhin et al. [12] emphasized the importance of focusing on semantics through dense embeddings in retrieval. Following this, Glass et al. [13] introduced a reranker that allows retrieval results whose scores are not directly comparable to be merged, enabling an ensemble of BM25 and neural retrieval. Luan et al. [14] examined the capacity of such dense dual-encoder representations, showing empirically that fixed-length encodings are limited in retrieving long documents, and explored sparse-dense hybrids to recover the precision of lexical matching. COIL [15] scores documents using the contextualized representations of tokens that overlap between query and document, while SPLADE [16] learns sparse term weights over the full vocabulary with an explicit expansion mechanism that mitigates vocabulary mismatch.

Such advancements in retrievers have opened up research on how retrieval should be conducted, along with applying these concepts in chatbots. Feng et al. [17] proposed a task and benchmark for grounding multiple documents in goal-oriented dialogues. Gao et al. [9] developed a benchmark that not only grounds information but also directly incorporates citations within the text for a more comprehensive evaluation.

2.2 Citation in LLMs

Citation methods using LLMs can be broadly categorized into non-parametric and parametric approaches [18]. The non-parametric approach is divided into pre-hoc and post-hoc citations. Pre-hoc citation involves identifying the need for citations before response generation. This method utilizes an information retrieval (IR) system to source relevant external information, which is then integrated into the response to enhance accuracy and reference quality. On the other hand, the post-hoc strategy involves generating a response first and then reviewing it to identify areas where citations may be necessary [9, 19]. In this case, an IR system is employed to retrieve relevant content

for citation integration, ensuring the response is well-supported and factually grounded.

Traditional post-hoc citation methods primarily rely on dense embedding similarity search, which matches generated text with retrieved documents using inner-product calculations. However, these methods often fail to capture specific keyword relevance. Our proposed method addresses this limitation by explicitly leveraging keyword overlap via Jaccard similarity, leading to higher precision and recall. We note that our method is not a sparse or lexical retrieval model, and differs from that line of work in four respects. First, the task differs: sparse retrieval such as BM25 [20] ranks documents for a query, whereas our method assigns citations to individual sentences of an already generated response over a fixed set of retrieved documents. Second, the signal is selected before matching: rather than weighting all tokens, we first extract domain-specific keywords (Table 3), a choice grounded in the keyword-retention observation of Sec. 3. Third, the computation is set-based: unlike BM25 and TF-IDF [21], which weight terms by frequency and inverse document frequency, we use only the intersection and union of keyword sets, so no corpus statistics or inverted index are required. Fourth, the computation requires no neural training or inference, which yields the efficiency reported in Fig. 3 and makes the matched keyword set directly inspectable (Fig. 1).

Habibi and Popescu-Belis [22] suggested the method of extracting multiple queries from a set of keywords and using these to recommend related English Wikipedia articles. Li et al. [23] showed the use of keywords to enhance the performance of abstractive sentence summarization in sequence-to-sequence models.

Second, Xiong et al. [24] performed TextRank [25] on news data by utilizing the semantic information of keywords for clustering. There is a study conducting information retrieval in keyword-based datasets like news by applying keyword extraction [26].

3 Motivations

The concept of a “keyword”, as defined by Hulth [27], refers to a word that captures the essence of a document’s content. In the field of NLP, keywords play a pivotal role in discerning user intent and identifying the central theme of a text because extracting keywords can reflect the semantics of a sentence [28]. This understanding led us to hypothesize that a generative model is more likely to use a keyword in its original form, which could result in its repeated occurrence in the generated text.

To validate this hypothesis, we conducted an experiment using a RAG approach with various models, including Llama2-7B Chat, Vicuna-7B, Mistral-7B, and GPT-4. The experiment involved generating responses to datasets from ASQA, QAMPARI, and MedDialog [29], followed by measuring the keyword overlap ratio, which means Jaccard similarity between extracted keyword in the generated responses and those in the retrieved documents to understand how these models handle and integrate keywords in their outputs.

For the experiment, we first extracted the keywords and stemmed them to their root forms. After this sequence, we calculated the probability of words that exist in the document reappearing in the generated sentences. As a control, we computed the same overlap ratio for non-keyword general nouns, since keywords are typically nouns. Fig. 4 illustrates these experiments.

The results observed across all models were consistent with our expectations, showing a significantly higher keyword overlap ratio compared to the general noun overlap ratio. Specifically, the average keyword overlap ratio in ASQA dataset was 0.21, compared to an average general noun overlap ratio of 0.08. In QAMPARI dataset, these ratios were 0.12 for keywords and 0.05 for general nouns, and MedDialog dataset showed an average keyword overlap ratio of 0.10 and a general noun overlap ratio of 0.06.

These results support our hypothesis: sentences generated by the LLM contain a consistently higher proportion of keywords present in the source documents than general nouns. This keyword retention can therefore serve as a signal for tracking the lineage of generated content, which motivates the keyword-based citation method described below.

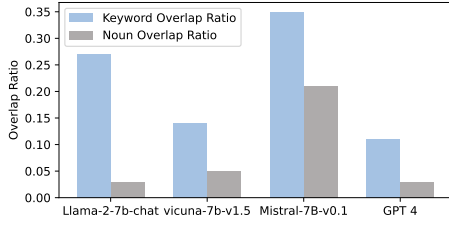
4 Method

To enhance sentence-by-sentence citations, our methodology, as depicted in Fig. 5, involves a comprehensive process encompassing keyword extraction, similarity computation, and citation generation with RAG pipeline.

Initially, a given query (denoted as x) is transformed by an encoder within a typical RAG structure. The response to the query, denoted as y , is composed of a collection of sentences, represented as $S = \{s_0, s_1, \dots, s_i\}$ and is supported by retrieval from a vector database, which provides external sources, represented as $D = \{d_0, d_1, \dots, d_k\}$. We employ a fine-tuned BERT model (as detailed in Appendix A) to extract keywords from the sentence set (denoted as K_S) and keywords from the document set (denoted as K_D). The extractor is selected by domain: BioBERT-based disease, genetic, and chemical NER models for MedDialog, and a general-purpose keyword extractor for ASQA and QAMPARI (Table 3), so that the extracted terms match the vocabulary of each corpus. After finding keywords, the keywords are stemmed for extracting the root forms of the words. This not only helps in standardizing keywords from diverse syntactic variations but also optimizes the matching process during the similarity calculations.

For the similarity computation, we calculate the Jaccard similarity for these keywords and create a similarity matrix between S and D to facilitate citations.

The Jaccard similarity, a measure of keyword overlap between S and D , is calculated by assessing the intersection and union of K_S and K_D . The formula for calculating Jaccard similarity between the keywords of



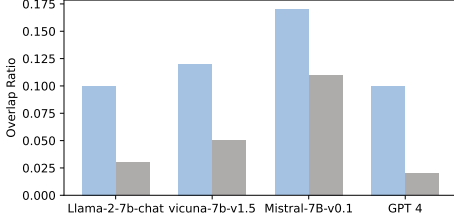
Generated Output

Ethiopia won the war between Ethiopia and Italy in 1935-1937.

Provided Context

The Second Italo-Ethiopian War, also referred to as the Second Italo-Abyssinian War, was a war of aggression which was fought between Italy and Ethiopia from October 1935 to February 1937.

(a) ASQA



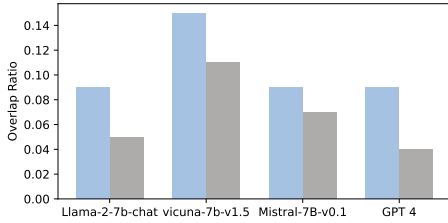
Generated Output

[...] Mani Ratnam wrote the script for and served as producer of the movie "Mouna Ragam" (1986).

Provided Context

[...] known professionally as Mani Ratnam, is an Indian film director, film producer and screenwriter ... his romantic trilogy of Mouna Ragam (1986).

(b) QAMPARI



Generated Output

[...] it seems that the patient had a novel bacterial species isolated from their rectal swab, which was identified as Pseudocitrobacter limerickensis.

Provided Context

[...] a novel bacterial species was isolated, based on rectal swab screening [...] phylogenetic analysis confirmed identification as Pseudocitrobacter [...]

(c) MedDialog

Figure 4: Comparison of keyword and noun (non-keyword) overlap ratios across four different models (Llama2-7b-chat, Vicuna-7b, Mistral-7b, GPT-4) and three datasets (ASQA, QAMPARI, MedDialog) where the subfigures are arranged in the order of ASQA, QAMPARI, and MedDialog, respectively. The words highlighted in the same color are the overlapping keywords between the retrieved document and the generated sentence.

the sentence K_{S_i} and the document K_{D_j} is as follows:

$$J(K_{S_i}, K_{D_j}) = \frac{|K_{S_i} \cap K_{D_j}|}{|K_{S_i} \cup K_{D_j}|}$$

Here, $|K_{S_i} \cap K_{D_j}|$ represents the number of keywords common between the sentence and the document, while $|K_{S_i} \cup K_{D_j}|$ represents the total number of unique keywords present in both the sentence and the document. The resulting score matrix, denoted as M is defined as:

$$M_{ij} = J(K_{S_i}, K_{D_j})$$

where each element M_{ij} represents the Jaccard similarity score between the i -th sentence and the j -th document. For each sentence s_i , the document d_j with the highest similarity score above a set threshold is chosen for citation. We set the threshold as 0.2. To justify this value, we examined the score distribution over the 200 generated responses used in Appendix C.1 (3,876 sentence-document pairs). 63.8% of the pairs receive a score of exactly zero and only 16.0% fall strictly between 0 and 0.2, so scores are concentrated away from the decision boundary rather than spread across it. Sweeping the threshold (Table 5) shows that citation decisions — whether a sentence is cited, and which document is selected — remain stable between 0.15 and 0.25, where

at least 93.8% of decisions are identical to those at 0.2. Stability degrades outside this range (87.5% at 0.30 and 74.7% at 0.50). We therefore set the threshold at the center of this stable region. The citation information, including relevant bibliographic data and URLs, is then integrated into the output text, directly linking each sentence to its source.

In addition, the reason for choosing Jaccard similarity instead of similar algorithms like ROUGE or BLEU was that it offered the fastest speed among other algorithms, as shown in Appendix C.1. We conducted a comparative analysis of processing times for various similarity and sequence matching algorithms, including Jaccard Similarity, BLEU-1, BLEU-2, ROUGE-1, ROUGE-2, Longest Common Subsequence, and TF-IDF as shown in Fig. 6. We processed 200 generated responses using each method to evaluate their processing time. The results indicated that Jaccard Similarity was the fastest, requiring only 0.045 seconds for 200 generated responses among the similar algorithms.

Our method showed its versatility by handling chitchat sentences, which usually contain fewer keywords. Unlike other citation methods that require explicit mechanisms to determine the necessity of a citation or to identify chitchat [30, 31], our approach in-

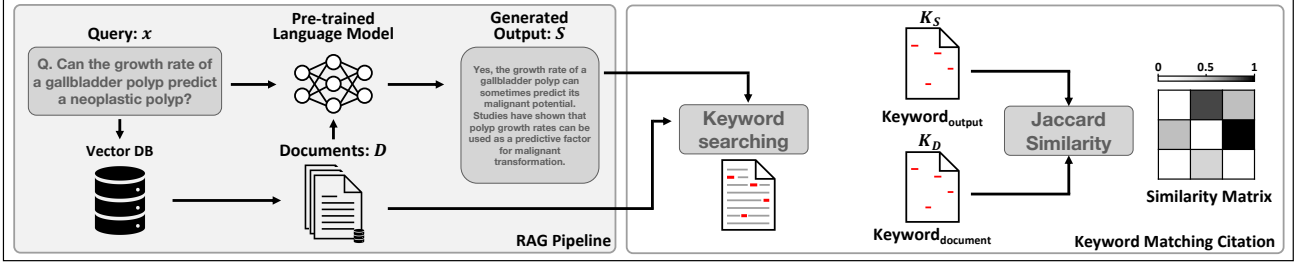


Figure 5: Illustration of our citation overall framework

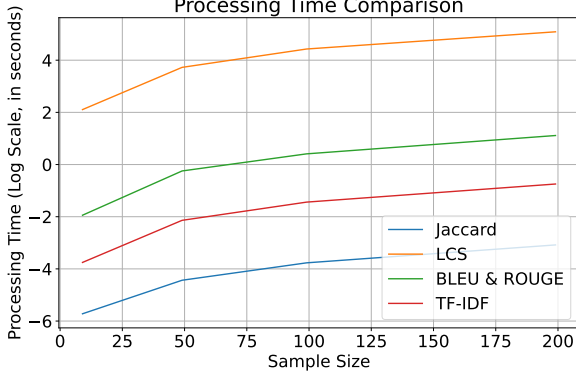


Figure 6: Comparison of processing time in text similarity and sequence matching algorithms

Algorithm 1: Calculate similarity scores using Jaccard Index

Input : LLM, RAG tool, N number of sentences

Function JaccardScore(sentence, document):
 $kw_s \leftarrow \text{extract_keyword}(\text{sentence});$
 $kw_d \leftarrow \text{extract_keyword}(\text{document});$
 $\text{return jaccard}(kw_s, kw_d);$

Function KeywordCitation(LLM, RAG, query):
 $\text{docs} \leftarrow \text{RAG}(\text{query});$
 $\text{sentences} \leftarrow \text{LLM}(\text{docs}, \text{query});$
 $N \leftarrow |\text{sentences}|;$
 $M \leftarrow |\text{docs}|;$
 $\text{score} \leftarrow \text{initialize matrix of size } N \times M;$
for s_i **in** sentences **do**
 for d_j **in** docs **do**
 $\text{score}[i, j] \leftarrow \text{JaccardScore}(s_i, d_j);$
 $\text{selected} \leftarrow \text{argmax}(\text{score}[i, :]);$
 $\text{Cite}(\text{sentences}[i], \text{selected});$

herently exhibits a tendency not to attach citations to miscellaneous sentences with minimal keyword content. This feature highlights the effectiveness of our method in discerning the relevance and context of sentences for citation without relying on additional identifiers. In other words, phrases like chit-chat are not derived from existing retrieved documents but instead originate from the pre-trained knowledge of LLMs.

5 Empirical Experiments

In this section, we will cover descriptions of our task and dataset (as detailed in Sec. 5.1), the baseline model and experimental settings (as detailed in Sec. 5.2), and the

results of automatic evaluations (as detailed in Sec. 5.3).

5.1 Task and Dataset

Task Our study’s primary objective is to assess and enhance the accuracy of citations in LLMs. In the development of our study, we adopted ALCE framework, a state-of-the-art neural network-based citation framework introduced in, which utilizes dense embeddings from sentence transformers. We selected ALCE framework as our baseline due to its comprehensive use of various LLMs, implemented through a neural network. Additionally, to the best of our knowledge, ALCE is the only available benchmark for citation. Therefore, it serves as an adequate baseline by employing a variety of LLMs and a neural network.

Datasets Our approach is designed for broad applicability across various domains, as evidenced by our experiments with datasets such as ASQA and QAMPARI, which draw from a diverse range of wiki-based topics. We utilized ASQA [32] and QAMPARI [33] datasets from the ALCE [9] framework to evaluate citation quality. ALCE evaluates the citation using the Natural Language Inference (NLI) model, TRUE [34], to determine the entailment of the generated sentence and the retrieved documents. The NLI model outputs 1 when the two sentences exhibit an entailment relationship and 0 otherwise. ASQA is a wiki-based factoid dataset containing in-depth answers to relatively ambiguous questions. QAMPARI is a QA dataset that, in contrast to typical wiki-based open domain question answering tasks, collected and organized multiple answers for a single question.

Expandability in specialized domains Moreover, we extended our experiments to the medical domain, where keywords and citations are important. In the case of MedDialog [29], we increased the difficulty level compared to existing datasets, focusing on evaluating citations within dialogues. We also collected medical documents and evaluated our framework using the RAG pipeline, demonstrating adaptability in the medical domain. To address this, we manually collected PubMed and medical-related documents, converting them into vector databases for the RAG pipeline. We conducted post-hoc citations on the doctor’s responses in the generated dialog. The increased difficulty arises from the inclusion of phrases, such as chit-chat, that do not require citation. So MedDialog presents a more complex

Table 1: Recall, precision, and GPT-4 evaluation of citation quality on ASQA, QAMPARI, and MedDialog for models up to Llama-13B. The baseline model scores are extracted from the ALCE-reported results.

Model	Method	ASQA			QAMPARI			MedDialog
		Recall	Precision	F1	Recall	Precision	F1	GPT-4
Llama2-7B Chat	Baseline	50.9	47.5	49.1	10.6	10.9	10.7	2.09
	Keyword Matching	56.1	49.7	52.7	11.0	11.1	11.0	2.58
Llama2-13B Chat	Baseline	38.4	39.4	38.9	9.6	9.7	9.6	2.03
	Keyword Matching	52.8	49.6	51.1	10.7	10.6	10.6	2.60
Vicuna-7B	Baseline	40.3	42.6	41.4	10.1	10.9	10.5	2.50
	Keyword Matching	50.5	49.2	49.8	13.1	13.2	13.1	2.82
Llama-7B	Baseline	6.2	9.2	7.4	5.1	5.7	5.4	1.5
	Keyword Matching	28.1	28.1	28.1	7.3	7.7	7.5	1.3
Llama-13B	Baseline	10.6	15.4	12.6	6.7	7.1	6.9	2.28
	Keyword Matching	29.5	29.3	29.4	8.7	8.8	8.7	2.69

case with its conversational nature and specialized terminology. This evidences our methodology’s capacity to transcend the medical domain, offering versatile solutions across different fields.

5.2 Experimental Setup

Baseline model Our baseline is the ALCE framework, and it is a benchmark created by applying both pre-hoc and post-hoc citations to ASQA, QAMPARI, and etc, with various LLMs. The pre-hoc citation is generated during the LLM generation step with a few shot examples. Then post-hoc citation is processed by using the sentence transformer ‘gtr-t5-xxl’ model and each generated sentence is embedded using the model. This embedding process is used for making a similarity matrix through maximum inner product search (matrix multiplication) between the embedded sentence matrix and the Wikipedia documents matrix. The model ranks the top- k Wikipedia documents that the sentence will be cited.

As shown in Table 1 and Table 4, we conducted an ablation study to demonstrate that our model not only performs well with the five models recorded in the ALCE paper but also excels with a variety of other LLMs such as Llama-13B, and Mistral-7B-Instruct. Cases not mentioned in the ALCE paper like Table 4 are in Appendix C.

Experimental setting We follow the default settings of ALCE to evaluate precision and recall metrics for ASQA and QAMPARI. We also performed GPT-4 evaluation as mentioned in Sec. 5.3 on ASQA, QAMPARI, and MedDialog. For inference, we used 4× Nvidia 3090 GPUs and 4× Nvidia A6000 GPUs. The evaluation sets consist of 100 instances each for ASQA and QAMPARI, and 61 for MedDialog, the latter being the entire test set released on HuggingFace. Given these sample sizes, we report the results as consistent trends across nine models (Table 1 and 4) rather than as precise performance margins. Further details of experiments are in Appendix A.

Metrics The performance of the models was evaluated via automatic evaluation using the ASQA, QAMPARI, and MedDialog datasets within the ALCE framework, which employs per-sentence citation. Recall and precision were used as evaluation metrics for citations. There is no ground truth for MedDialog datasets we adopted GPT-4 evaluations, a 4-point Likert scale for assessing citations using GPT-4 turbo [2], and the instructions for evaluation. We provided the LLM with two cited results for comparison, including the documents each cites. The baseline and our methods were anonymized, with cited sentences and referenced documents as inputs. Instructions were structured to evaluate based on common criteria. Details of GPT-4 evaluations are in Appendix B. We also conducted a human evaluation with the same criteria (as detailed in Sec. 5.4).

5.3 Automatic Evaluation and Results

Our approach demonstrated significant improvements across all datasets and models, surpassing the baseline models even in human evaluation as shown in Table 1 and Table 4. Results and details of human evaluation are in Sec. 5.4

In the ASQA dataset, improvements were notable regardless of the model’s initial performance. For example, Llama2-13B Chat saw a 14% absolute increase in recall and a 10% absolute increase in precision. At the same time, Llama-7B experienced a four times increase in recall and a three times improvement in precision compared to the baseline. QAMPARI shows a different pattern. While recall and precision improve across models, the gains are markedly smaller than on ASQA, and in the human evaluation the baseline scores slightly higher than our method (2.52 vs. 2.28, Table 2). Notably, the GPT-4 evaluation and the human evaluation agree on this relative ordering, indicating that the result reflects a property of the dataset rather than a bias of either evaluation protocol. QAMPARI differs from the other datasets in that each question is paired with mul-

Table 2: Human evaluation on ASQA, QAMAPRI, and MedDialog on Llama2-7B Chat model

Dataset	ASQA	QAMPARI	MedDialog	Avg
Baseline	2.10	2.52	1.83	2.15
Keyword Matching	3.36	2.28	2.88	2.84

tuple answers collected from separate paragraphs (Sec. 5.1), and we discuss the implications of this structure for keyword-based matching.

For MedDialog, due to the absence of a definitive answer measurement framework, we adopted a GPT-4 evaluation score. Our method achieved an average increase of 0.26 points across the models, although the Llama-7B model underperformed relative to the baseline by 0.2 points. This discrepancy was mainly due to the incoherent sentence formation in less structured conversation by Llama-7B, which affected its GPT-4 evaluation scores.

Our method outperformed baseline model, even with less structured, domain-specific texts like MedDialog conversations. Our framework performed well in these scenarios, demonstrating its capability through extensive testing in complex environments. This demonstrates our method’s ability to generalize across both generic Wikipedia and specialized medical domains.

We also report the consistency of the improvement across models rather than instance-level confidence intervals. Treating each model in Table 1 and 4 as a paired observation, a two-sided sign test over the nine models gives $p = 0.004$ for ASQA F1 (9 of 9 improved), $p = 0.004$ for QAMPARI F1 (9 of 9), and $p = 0.039$ for the MedDialog GPT-4 score (8 of 9; the single exception is Llama-7B, discussed above). The improvement is therefore consistent across backbones rather than driven by a particular model.

Overall, our model demonstrated high and consistent performance not only on Wikipedia-based data for general domains such as ASQA and QAMPARI but also on the specialized domain, MedDialog in both quantitative and qualitative evaluations. Our framework has shown scalability across datasets, achieving high performance quickly with fewer resources, as illustrated in Fig. 3.

5.4 Human Evaluation

To confirm that our automatic evaluation aligns with human judgment we perform human evaluation on three datasets with Llama2-7B Chat model. Human evaluation was performed by four evaluators, who received exactly the same prompt and criteria given to GPT-4 (Fig. 7, Fig. 8). The baseline and our method were anonymized as Model A and Model B, and each evaluator was shown the cited sentences together with the referenced documents. To reduce ambiguity between adjacent scores, each point on the 4-point Likert scale was defined by an explicit band of correctly cited content (below 25%, 25–50%, 50–75%, and 75% or above), and unnecessary citations in chit-chat contexts were penalized under the same rule for both methods. The

results are demonstrated in Table 2. Our method outperformed the baseline method on ASQA (3.36 compared to 2.10) and MedDialog (2.88 compared to 1.83). On QAMPARI, the baseline method outperformed our method by a small portion (2.52 vs. 2.28) however, on average, our method achieved a better score (2.84 compared to 2.15).

The alignment between the human evaluation results and GPT-4 evaluation, as shown in Table 1 and Table 4 and Table 2, is noteworthy. This consistency is particularly evident in the ASQA and MedDialog datasets, where both evaluations favored our method, and in the QAMPARI dataset, where both indicated a lower score for our method compared to the baseline method. This correlation between GPT-4 evaluation and human evaluation underscores the reliability of the GPT-4 evaluation in reflecting the effectiveness of our citation methodology.

6 Limitations&Future Work

Limitations First, matching operates on stemmed keyword forms (Sec. 4), so it aligns morphological variants but does not capture cases where the same concept is expressed in different terms — a common situation in the medical domain. Second, performance depends on the quality of the keyword extractor; we selected extractors suited to each domain (Table 3) but did not quantify the method’s sensitivity to this choice. Third, the length analysis in Appendix C.4 covers the range of retrieved passages and does not establish behaviour on substantially longer documents. Fourth, our comparison against lexical approaches is limited to processing time (Appendix C.1); we did not evaluate citation accuracy against sparse retrieval baselines.

Future work We did not use dense embedding and training, so we aim to integrate the advantages of both dense embeddings and keyword-centric approaches. Re2G [13] shows that retrieval results from a lexical retriever and a neural retriever can be combined even when their scores are not directly comparable, which supports the feasibility of combining dense and keyword-based signals. We note, however, that Re2G obtains this combination through a trained reranker, whereas preserving the training-free property and the efficiency reported in Fig. 3 would require a combination rule that needs no training. As for the limitation on term variation discussed above, expansion-based sparse models such as SPLADE [16] address vocabulary mismatch directly by adding terms that do not appear in the document, and adapting such expansion to keyword-based citation is a further direction. By adjusting the training rewards to focus on keywords, as suggested by Asai et al. [35], we anticipate an enhancement in the relevance and accuracy of the generated citations.

7 Conclusion

With the development of LLMs and RAG, research about citation has become necessary. We have demonstrated a method that is more accurate, faster and requires fewer resources by using a keyword-centric approach. When utilizing RAG, LLMs tend to reproduce keywords as we experimented. By creating a similarity matrix using Jaccard similarity focused on keywords, we have made it possible to achieve this without the need for training complex neural networks or using traditional inner-product-based methods. To demonstrate the superiority of our algorithm, we showed that it performs well in the general domain using the ALCE framework, exhibited higher performance on the ASQA and QAMPARI datasets compared to conventional methods, and achieved high performance in the medical field, which uses specialized vocabulary. Therefore, this research reveals the substantial potential of post-hoc citations in improving the reliability of LLM-generated content.

References

- [1] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.
- [2] OpenAI. Gpt-4 technical report, 2023.
- [3] Brendan Luyt and Daniel Tan. Improving wikipedia’s credibility: References and citations in a sample of history articles. *Journal of the American Society for Information Science and Technology*, 61(4):715–722, 2010.
- [4] CJ Moerman and HB Bueno-de Mesquita. The epidemiology of gallbladder cancer: lifestyle related risk factors and limited surgical possibilities for prevention. *Hepato-gastroenterology*, 46(27):1533–1539, 1999.
- [5] Raymond Diep, Paul Lombardo, and Michal Schneider. The growth rates of solitary gallbladder polyps compared to multi-polyps: a quantitative analysis. *Australasian Journal of Ultrasound in Medicine*, 25(1):28–35, 2022.
- [6] Ji Won Han, Young Hoon Choi, In Seok Lee, Ho Joong Choi, Tae Ho Hong, and Young Kyoung You. Gallbladder polyps growth rate is an independent risk factor for neoplastic polyps. *United European Gastroenterology Journal*, 10(7):651–656, 2022.
- [7] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33:9459–9474, 2020.
- [8] Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. Pubmedqa: A dataset for biomedical research question answering. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2567–2577, 2019.
- [9] Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. *arXiv preprint arXiv:2305.14627*, 2023.
- [10] Baolin Peng, Michel Galley, Pengcheng He, Hao Cheng, Yujia Xie, Yu Hu, Qiuyuan Huang, Lars Liden, Zhou Yu, Weizhu Chen, et al. Check your facts and try again: Improving large language models with external knowledge and automated feedback. *arXiv preprint arXiv:2302.12813*, 2023.
- [11] Nouha Dziri, Andrea Madotto, Osmar Zaiane, and Avishek Joey Bose. Neural path hunter: Reducing hallucination in dialogue systems via path grounding. *arXiv preprint arXiv:2104.08455*, 2021.
- [12] Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. *arXiv preprint arXiv:2004.04906*, 2020.
- [13] Michael Glass, Gaetano Rossiello, Md Faisal Mahbub Chowdhury, Ankita Rajaram Naik, Pengshan Cai, and Alfio Gliozzo. Re2g: Retrieve, rerank, generate. *arXiv preprint arXiv:2207.06300*, 2022.
- [14] Yi Luan, Jacob Eisenstein, Kristina Toutanova, and Michael Collins. Sparse, dense, and attentional representations for text retrieval. *Transactions of the Association for Computational Linguistics*, 9:329–345, 2021. doi: 10.1162/tacl_a_00369. URL <https://aclanthology.org/2021.tacl-1.20/>.
- [15] Luyu Gao, Zhuyun Dai, and Jamie Callan. COIL: Revisit exact lexical match in information retrieval with contextualized inverted list. In Kristina Toutanova, Anna Rumshisky, Luke Zettlemoyer, Dilek Hakkani-Tur, Iz Beltagy, Steven Bethard, Ryan Cotterell, Tanmoy Chakraborty, and Yichao Zhou, editors, *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3030–3042, Online, June 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.naacl-main.241. URL <https://aclanthology.org/2021.naacl-main.241/>.
- [16] Thibault Formal, Benjamin Piwowarski, and Stéphane Clinchant. Splade: Sparse lexical and

- expansion model for first stage ranking. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR '21*, page 2288–2292, New York, NY, USA, 2021. Association for Computing Machinery. ISBN 9781450380379. doi: 10.1145/3404835.3463098. URL <https://doi.org/10.1145/3404835.3463098>.
- [17] Song Feng, Siva Sankalp Patel, Hui Wan, and Sachindra Joshi. Multidoc2dial: Modeling dialogues grounded in multiple documents. *arXiv preprint arXiv:2109.12595*, 2021.
- [18] Jie Huang and Kevin Chen-Chuan Chang. Citation: A key to building responsible and accountable large language models. *arXiv preprint arXiv:2307.02185*, 2023.
- [19] Bernd Bohnet, Vinh Q Tran, Pat Verga, Roei Aharoni, Daniel Andor, Livio Baldini Soares, Massimiliano Ciaramita, Jacob Eisenstein, Kuzman Ganchev, Jonathan Herzig, et al. Attributed question answering: Evaluation and modeling for attributed large language models. *arXiv preprint arXiv:2212.08037*, 2022.
- [20] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: Bm25 and beyond. *Found. Trends Inf. Retr.*, 3(4):333–389, April 2009. ISSN 1554-0669. doi: 10.1561/15000000019. URL <https://doi.org/10.1561/15000000019>.
- [21] Gerard Salton and Christopher Buckley. Term-weighting approaches in automatic text retrieval. *Inf. Process. Manage.*, 24(5):513–523, August 1988. ISSN 0306-4573. doi: 10.1016/0306-4573(88)90021-0. URL [https://doi.org/10.1016/0306-4573\(88\)90021-0](https://doi.org/10.1016/0306-4573(88)90021-0).
- [22] Maryam Habibi and Andrei Popescu-Belis. Keyword extraction and clustering for document recommendation in conversations. *IEEE/ACM Transactions on audio, speech, and language processing*, 23(4):746–759, 2015.
- [23] Haoran Li, Junnan Zhu, Jiajun Zhang, Chengqing Zong, and Xiaodong He. Keywords-guided abstractive sentence summarization. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(05):8196–8203, Apr. 2020. doi: 10.1609/aaai.v34i05.6333. URL <https://ojs.aaai.org/index.php/AAAI/article/view/6333>.
- [24] Ao Xiong, Derong Liu, Hongkang Tian, Zhengyuan Liu, Peng Yu, and Michel Kadoch. News keyword extraction algorithm based on semantic clustering and word graph model. *Tsinghua Science and Technology*, 26(6):886–893, 2021.
- [25] Rada Mihalcea and Paul Tarau. TextRank: Bringing order into text. In Dekang Lin and Dekai Wu, editors, *Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing*, pages 404–411, Barcelona, Spain, July 2004. Association for Computational Linguistics. URL <https://aclanthology.org/W04-3252>.
- [26] Oleg Borisov, Mohammad Aliannejadi, and Fabio Crestani. Keyword extraction for improved document retrieval in conversational search. *arXiv preprint arXiv:2109.05979*, 2021.
- [27] Anette Hulth. Improved automatic keyword extraction given more linguistic knowledge. In *Proceedings of the 2003 conference on Empirical methods in natural language processing*, pages 216–223, 2003.
- [28] Prafull Sharma and Yingbo Li. Self-supervised contextual keyword and keyphrase retrieval with self-labelling. 2019.
- [29] Guangtao Zeng, Wenmian Yang, Zeqian Ju, Yue Yang, Sicheng Wang, Ruisi Zhang, Meng Zhou, Jiaqi Zeng, Xiangyu Dong, Ruoyu Zhang, et al. Meddialog: Large-scale medical dialogue datasets. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9241–9250, 2020.
- [30] Hanan Aljuaid, Rimsha Iftikhar, Shahbaz Ahmad, Muhammad Asif, and Muhammad Tanvir Afzal. Important citation identification using sentiment analysis of in-text citations. *Telematics and Informatics*, 56:101492, 2021.
- [31] Giovanni Colavizza, Iain Hrynaskiewicz, Isla Staden, Kirstie Whitaker, and Barbara McGillivray. The citation advantage of linking publications to research data. *PloS one*, 15(4): e0230416, 2020.
- [32] Ivan Stelmakh, Yi Luan, Bhuwan Dhingra, and Ming-Wei Chang. ASQA: Factoid questions meet long-form answers. In Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang, editors, *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 8273–8288, Abu Dhabi, United Arab Emirates, December 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.566. URL <https://aclanthology.org/2022.emnlp-main.566>.
- [33] Samuel Joseph Amouyal, Tomer Wolfson, Ohad Rubin, Ori Yoran, Jonathan Herzig, and Jonathan Berant. Qampari: An open-domain question answering benchmark for questions with many answers from multiple paragraphs, 2023.
- [34] Or Honovich, Roei Aharoni, Jonathan Herzig, Hagai Taitelbaum, Doron Kukliansy, Vered Cohen, Thomas Scialom, Idan Szpektor, Avinatan Hassidim, and Yossi Matias. TRUE: Re-evaluating factual consistency evaluation. In Marine Carpuat, Marie-Catherine de Marneffe, and Ivan Vladimir

Meza Ruiz, editors, *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3905–3920, Seattle, United States, July 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.naacl-main.287. URL <https://aclanthology.org/2022.naacl-main.287>.

- [35] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-rag: Learning to retrieve, generate, and critique through self-reflection. *arXiv preprint arXiv:2310.11511*, 2023.
- [36] Sunjun Kweon, Junu Kim, Jiyou Kim, Sujeong Im, Eunbyeol Cho, Seongsu Bae, Jungwoo Oh, Gyubok Lee, Jong Hak Moon, Seng Chan You, et al. Publicly shareable clinical large language model built on synthetic clinical notes. *arXiv preprint arXiv:2309.00237*, 2023.

Table 3: Illustration of models used for keyword extraction for various datasets

Dataset	NER Module
MedDialog	alvaroalon2/biobert_diseases_ner ⁴
	alvaroalon2/biobert_genetic_ner ⁵
	alvaroalon2/biobert_chemical_ner ⁶
ASQA QAMPARI	yanekyuk/bert-uncased-keyword-extractor ⁷

A Experimental settings

The number of tests on ASQA and QAMPARI is 100, and on MedDialog is 61, due to the number of test sets released on MedDialog in HuggingFace being 61. The temperature for GPT-4 evaluation is set at 0.7. The keyword extractor model is detailed in Table 3.

For ASQA and QAMPARI, we followed the ALCE instructions, such as using the top k retrieved documents. In the case of MedDialog, we used the top 15 retrieved documents. The retrieved documents of MedDialog are from Pubmed and Medline. We embedded these documents and utilized them for RAG.

We conducted an ablation study to evaluate our citation method using OPT-6.7B⁸, GPT-J 6.7B⁹, Llama-13B,¹⁰ Llama2-13B¹¹, Vicuna-7B¹², and Mistral-7B-Instruct¹³ as shown in Table 1 and Table 4.

⁴https://huggingface.co/alvaroalon2/biobert_diseases_ner

⁵https://huggingface.co/alvaroalon2/biobert_genetic_ner

⁶https://huggingface.co/alvaroalon2/biobert_chemical_ner

⁷<https://huggingface.co/yanekyuk/bert-uncased-keyword-extractor>

⁸<https://huggingface.co/facebook/opt-6.7b>,

⁹<https://huggingface.co/ElleutherAI/gpt-j-6b>

¹⁰<https://huggingface.co/huggyllama/llama-13b>

¹¹<https://huggingface.co/meta-llama/Llama-2-13b-hf>

¹²<https://huggingface.co/lmsys/vicuna-7b-v1.5>

¹³<https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.1>

You are an intelligent expert.

Your task is to evaluate the quality of the citations based on the following criteria.

[The start of the response of Model A]

[the response of Model A]

[The end of the response of Model A]

[The start of the retrieved documents of Model A]

[the retrieved documents of Model A]

[The end of the retrieved documents of Model A]

[The start of the response of Model B]

[the response of Model B]

[The end of the response of Model B]

[The start of the retrieved documents of Model B]

[the retrieved documents of Model B]

[The end of the retrieved documents of Model B]

Instruction

Cited result A and cited result B will be provided. your task is to evaluate each of them separately based on the following criteria. Additionally, you are required to suggest a total score for each model, indicating them as ‘Total Score A: [score], Total Score B: [score]’.

Another important point is please evaluate generously and leniently. However, good scores should only be awarded for citations that are included in essential sections or sentences highly relevant to the retrieved document, not for indiscriminate citations. It means that not every sentence requires a citation. For instance, in casual conversations like chit-chat, it is appropriate not to include citations. Incorrectly citing in unnecessary situations is wrong, while not citing is correct. For example, in everyday conversations, such as simply expressing gratitude or asking questions, citations should not be made. For everyday dialogue, such as expressing thanks or posing questions, citations should be avoided. Scores should be reduced for unnecessary citations and increased when they are appropriately omitted.

Additionally, in cases where answers can be given based on general knowledge without using retrieved documents, it’s not necessary to cite. In these situations, not citing is the correct approach. Lastly, answers that don’t include any citations at all are judged as not requiring citations for any sentences, so scores should be assigned accordingly.

Criteria

[The start of criteria]

[Criteria]

[The end of the Criteria]

Figure 7: Instruction prompt example

B Prompts

In this section, we detailed the prompts used for GPT-4 evaluation in our MedDialog experiments.

Evaluation prompt Fig. 7 illustrates the entire sequence of both GPT-4 and human evaluation processes. We provided identical prompts and evaluation criteria to both GPT-4 and human annotators.

The instruction starts with a system prompt and an initial prompt, offering fundamental information about the task. Then, anonymous responses and retrieved documents are presented for the baseline citation method and our keyword-matching citation method, respectively. Subsequently, the remaining instructions and criteria are provided. Fig. 8 displays the common criteria used for both GPT-4 and human evaluations.

GPT-4 evaluation metric Appendix B shows our newly proposed evaluation metric, a 4-point Likert scale for assessing citations using GPT-4 turbo [2], and the instructions for evaluation. Previously, Kweon et al. [36] utilized a 4-point Likert scale using GPT-4 to evaluate synthetic clinical notes generated by models. Adapting this approach, we provide the LLM with two cited results for comparison, including the documents each result cites as input. We anonymized the baseline method and our method as model A and model B and provided cited sentences and the referenced documents as inputs. Subsequently, we structured the instructions to evaluate based on a common set of criteria.

A distinctive aspect of our approach is adding objective criteria. This is because using a 4-point scale can lead to ambiguous interpretations. By clearly defining how these objective criteria should be interpreted, LLM can enable more precise scoring. Furthermore, this approach ensured that the ranges corresponding to each score were clearly defined, preventing any particular score from being disproportionately represented. For instance, in cases where it is difficult to clearly distinguish between adjacent scores, such as 2 and 3, we assigned a score of 2 for responses with approximately 25% to 50% accurate citations for the response and a score of 3 for those with 50% to 75% accuracy.

C Other Experimental Results

C.1 Processing time ablation

We conducted a comparative analysis of processing times for various similarity and sequence matching algorithms, including Jaccard Similarity, BLEU-1, BLEU-2, ROUGE-1, ROUGE-2, Longest Common Subsequence (LCS), and TF-IDF as shown in Fig. 6. We processed 200 generated responses using each method to evaluate their processing time.

The results indicated that Jaccard Similarity was the fastest, requiring only 0.045 seconds for 200 generated responses, followed by TF-IDF, which took 0.467 seconds. In contrast, LCS was the most time-consuming,

1. **Unacceptable** (1 point): When the overall response correctly cites the document less than 25%, the overall grade is 1. This means that the model’s response does not contain any part of the cited document and cannot be answered using that document at all. And too many unnecessary citations were made in contexts like chitchat.
2. **Poor** (2 points): When the overall response correctly cites the document at least 25% but less than 50%, the overall grade is 2. This means that the model’s response can be inferred from the cited source, but the connection is unclear due to many insignificant contents. Another document might provide a better explanation. And many unnecessary citations were made in contexts like chitchat.
3. **Satisfactory** (3 points): When the overall response correctly cites the document at least 50% but less than 75%, the overall grade is 3. This means that the model’s response can be inferred from the cited source. However, it omits minor or insignificant content that the instruction requires. And few unnecessary citations were made in contexts like chitchat.
4. **Excellent** (4 points): When the overall response correctly cites the document 75% or more, the overall grade is 4. This means that the cited source clearly explains the model’s response, more so than other retrieved documents. And almost no unnecessary citations were made in contexts like chitchat.

Figure 8: GPT-4 and human evaluation criteria

Table 4: Recall, precision, and GPT-4 evaluation of citation quality on ASQA, QAMAPARI, and MedDialog for models after Llama-13B. The baseline model scores which are indicated by the symbol ^(*) of Llama2-13B, Mistral-7B-Instruct, OPT-6.7B, and GPT-J 6.7B are not reported within the paper of the ALCE framework. We reproduced these models within the ALCE framework for the ablation study. The symbol ^(•) indicates that the experiments were conducted with the top 5 documents, accommodating the smaller maximum context length of these models.

Model	Method	ASQA			QAMPARI			MedDialog
		Recall	Precision	F1	Recall	Precision	F1	GPT-4
Llama2-13B^(*)	Baseline	27.7	28.0	27.8	8.4	8.6	8.5	1.98
	Keyword Matching	33.2	31.2	32.2	8.9	8.9	8.9	2.23
Mistral-7B-Instruct^(*)	Baseline	22.8	27.1	24.8	9.8	10.4	10.1	2.11
	Keyword Matching	50.6	49.2	49.9	13.9	14.0	13.9	2.21
OPT-6.7B^(*)	Baseline	6.5	10.6	8.1	4.9	5.4	5.1	1.58 ^(•)
	Keyword Matching	23.6	23.5	23.5	6.6	6.7	6.6	1.64 ^(•)
GPT-J 6.7B^(*)	Baseline	9.6	13.3	11.2	6.8	7.6	7.2	1.70 ^(•)
	Keyword Matching	25.1	24.8	24.9	8.5	8.6	8.5	2.07 ^(•)
Average	Baseline	23.6	25.9	24.7	8.0	8.4	8.2	1.97
	Keyword Matching	38.8	37.1	37.9	9.8	9.9	9.8	2.23

Table 5: Effect of the citation threshold, measured on the 200 generated responses of Appendix C.1 (3,876 sentence–document pairs). A decision is counted as identical when both the citation/no-citation outcome and the selected document match those obtained at the threshold of 0.2.

Threshold	Cited sentences	Decisions identical to 0.2
0.05	52.8%	87.4%
0.10	50.4%	89.9%
0.15	44.4%	95.9%
0.20	40.2%	100.0%
0.25	34.1%	93.8%
0.30	27.8%	87.5%
0.40	19.0%	78.7%
0.50	15.0%	74.7%

taking 168.674 seconds. The processing times for the other methods were relatively similar, with BLEU-1 taking 2.881 seconds, BLEU-2 taking 2.882 seconds, and both ROUGE-1 and ROUGE-2 taking 2.875 seconds each.

In Fig. 6, BLEU-1, BLEU-2, ROUGE-1, and ROUGE-2 are grouped together due to their nearly identical processing times. The times are represented on a logarithmic scale for a clearer and more concise visualization of the comparative performance of these methods.

C.2 Cases not mentioned in the ALCE paper

Cases not mentioned in the ALCE paper are marked with an asterisk ^(*). For OPT-6.7B and GPT-J 6.7B, a bullet ^(•) was used to indicate that experiments were conducted only with the top 5 documents due to the context length limitations of the LLMs.

Similar to the overall trend, models not recorded in [9] also showed significantly improved performance

compared to before. In particular, models like Mistral-7B-Instruct, OPT-6.7B, and GPT-J 6.7B exhibited more than double the performance in the ASQA dataset. Vicuna-7B’s recall increased by 10% and precision by 6%, while OPT-6.7B saw a 3.5-times increase in recall and a two-times increase in precision. This improvement is also evident in the average performance

C.3 Threshold sensitivity

We set the citation threshold to 0.2 in Sec. 4. To examine how sensitive the method is to this choice, we recomputed the Jaccard score matrices for the 200 generated responses used in Appendix C.1, giving 3,876 sentence–document pairs. The scores are concentrated away from the decision boundary: 63.8% of the pairs receive a score of exactly zero, only 16.0% fall strictly between 0 and 0.2, and the remaining 20.3% lie at or above 0.2. Table 5 sweeps the threshold and reports how often the citation decision — both the citation/no-citation outcome and the selected document — is identical to the decision made at 0.2. Decisions are stable between 0.15 and 0.25, where at least 93.8% of them are unchanged, and degrade outside this range.

C.4 Effect of document length

Because our score is a set-based Jaccard index, a longer document could in principle enlarge the union and dilute the score. We examined this on the 800 retrieved documents of the Appendix C.1 set, which range from 4 to 174 words (median 142). Document length correlates moderately with the size of the extracted keyword set (Pearson $r = +0.279$, $p < 10^{-15}$), but shows essentially no relationship with the resulting Jaccard score (Pearson $r = -0.058$, $p = 0.099$; Spearman $r = -0.003$, $p = 0.92$). As Table 6 shows, the mean score varies only between 0.097 and 0.124 across length quartiles with no monotonic trend, and the share of citations assigned to each quartile is likewise non-monotonic. Within this

Table 6: Jaccard score by document length quartile, over the 800 retrieved documents of the Appendix C.1 set. $|K_D|$ is the size of the extracted keyword set. Citation share is the proportion of all citations assigned to documents in that quartile.

Quartile	Words	n	$ K_D $	Jaccard	Citation share
Q1 (shortest)	4–130	200	4.4	0.124	32.8%
Q2	130–142	179	6.9	0.097	15.1%
Q3	142–149	198	6.6	0.101	22.3%
Q4 (longest)	149–174	223	6.5	0.105	29.7%

range, keyword extraction absorbs most of the length variation before the similarity is computed. We did not test documents substantially longer than this range.